

2022 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces — Friction & Uncertainties

A puck (mass 0.350 kg) is released at speed 0.78 m s^{-1} and travels 2.160 m before stopping. (a) (i) Calculate the average acceleration, (ii) calculate the average force of friction. (b) The student suggests that using a more precise balance would best reduce the uncertainty in the average force. Explain why this statement is incorrect.

Worked solution

(a)(i) Average acceleration

Equation of motion: $v^2 = u^2 + 2as$

$$0^2 = (0.78)^2 + 2 \cdot a \cdot (2.160)$$

$$a = -(0.78^2) / (2 \times 2.160) = -0.141 \text{ m s}^{-2}$$

Answer: $a \approx -0.141 \text{ m s}^{-2}$

(a)(ii) Average friction force

$$F = m \cdot a = 0.350 \times (-0.141) = -0.049 \text{ N}$$

Magnitude of force: 0.049 N

(b) Reducing uncertainty

Mass of puck has only 0.3% uncertainty, which is very small.

Initial speed u has 1.3% uncertainty, and distance d has 0.05% uncertainty.

Since the calculation for force depends on u^2 and d , the uncertainty is dominated by the measurement of u .

Therefore, using a more precise balance to measure mass would not significantly reduce the overall uncertainty. Improving the measurement of u (initial speed) is much more effective.

Final answers

(a)(i) Average acceleration $\approx -0.141 \text{ m s}^{-2}$

(a)(ii) Magnitude of average force $\approx 0.049 \text{ N}$

(b) Incorrect: mass uncertainty is already very small; uncertainty in u dominates.

Revision tips

- Use $v^2 = u^2 + 2as$ for motion with constant acceleration.
- Negative acceleration means deceleration due to friction.
- Average friction force is $F = m \cdot a$.
- Compare percentage uncertainties to see which measurement dominates.

- To reduce overall uncertainty, improve the measurement with the largest percentage effect.